Jake M. Walker

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Biography

I was born in Edinburgh in 1997 and raised in St. Andrews, Scotland. I studied Mechanical Engineering at the University of Strathclyde (2015–2020). Towards the end of my studies, I became interested in engineering problems that cannot be addressed effectively through classical engineering approaches alone, e.g., due to computational complexity, limited understanding, or uncertainty in the available inputs. This is where I found my passion for using data-driven methods to the complement classical engineering approaches. After completing 1 year of research in this field, I pursued my Ph.D. at Delft University of Technology (2021–2025), during which I was the first author of more than 10 academic publications on the development and deployment of Artificial Intelligence (AI) tools for the optimal design and operation of maritime engineering assets. My experience spans both scarce- and big-data problems, including the creation of small datasets describing novel vessel topologies and their performance for AI-driven optimization pipelines, as well as the analysis of real operational time-series data from vessels, vessel subsystems, energy systems, and offshore wind turbines for digital twinning, condition monitoring, and forecasting. My expertise lies in solving engineering problems using AI and communicating the value of this research to both technical and wider audiences.

Experience

2025–Present **Researcher**, *De Voogt Naval Architects – Feadship*, Hoofddorp, Netherlands
I am responsible for identifying, developing, and implementing data-driven approaches within the Feadship organization to support the design, engineering, and construction of superyachts. My day-to-day activities include:

- Developing data-driven simulation approaches for onboard energy systems by modeling heating, cooling, and electrical demand onboard;
- Developing data-driven models of energy-system components including generators, batteries, heaters, and chillers to support the optimization of energy management systems;
- Performing operational-data analysis for sailing-profile characterization by combining heterogeneous data sources, including weather, metocean, AIS, and onboard data;
- Applying operational-data analysis methods to support hull and propeller-performance monitoring and identify changes in vessel performance over time;
- Leading a hull-form design optimization project funded by the RvO Maritime AI Innovation Lab, in partnership with MARIN, TU Delft, Damen, and C-Job;
- Roadmapping applications of AI-based technologies across the Feadship organization and presenting these opportunities to senior decision-makers, including C-level executives leading an organization of >3500 people.

2021–2025 **Ph.D. Candidate**, *Delft University of Technology*, Delft, Netherlands
My experience as a Ph.D. candidate demanded technical proficiency and project management skills. My research focused on designing a framework for optimizing vessel hull-forms using AI-based surrogate models. The key technical competencies included:

- Designing novel vessel topologies;
- Physical modeling of vessel designs using Computational Fluid Dynamics;
- Creating datasets describing shape variations and corresponding vessel KPIs;
- Using shallow and deep learning models to develop AI-based models;
- Embedding shallow and deep learning models into optimization workflows.

2020–2021 **Researcher**, *University of Strathclyde*, Glasgow, Scotland
I worked as a researcher in the Department of Naval Architecture, Ocean and Marine Engineering, contributing to data-driven and renewable-energy research.

Education

2021–2025 **Ph.D. Mechanical Engineering**, *Delft University of Technology*, Delft, Netherlands
Co-supervised by the Department of Computer Science at the *University of Genoa, Italy.*
Hull-Form Optimization using Computational Fluid Dynamics Data-Driven Surrogate Models
Promotors: Dr. Andrea Coraddu, Prof. Luca Oneto, & Prof. Bendiks-Jan Boersma

2022–2024 **Ph.D. Summer Schools**
April 2024 IEEE–Oceanic Engineering Society: Observing our Ocean, Singapore.
July 2024 4th TAILOR Summer School on Artificial Intelligence, Athens, Greece.
July 2023 DeepLearn 2023 Summer, Las Palmas, Gran Canaria.
July 2022 Theory and Practice of Learning from Data, Genoa, Italy.

2015–2020 **MEng Mechanical Engineering** (*with Distinction*), *University of Strathclyde*, Glasgow, Scotland

Teaching Experience

2025–Present **Lecturer – Research Skills**, *The Hague Executive Campus (THEC)*,
Den Haag, Netherlands
I designed and delivered a course on research skills for engineering professionals in the shipbuilding industry. The course introduced students to the conceptualisation, execution, documentation, and presentation of applied research projects. I developed lectures, workshop material, and assessment briefs for two Professional Product assignments and a boardroom assessment, supporting students in producing research outputs suitable for the programme's diploma requirements.

2022–2023 **Teaching Assistant - MT44000: Mechatronics**, *Delft University of Technology*, Delft, Netherlands
My responsibilities as a teaching assistant included assisting with the organization of lecture and tutorial material covering key concepts related to the design of autonomous vessel control systems. I was also responsible for the monitoring of the virtual classroom during the COVID-19 online period. I was required to be available to answer students' queries regarding class material.

2020–2022 **Teaching Assistant - NM836: Data Analysis for Engineering**,
University of Strathclyde, Glasgow, Scotland
My responsibilities as a teaching assistant included preparing & delivering weekly tutorial material covering key concepts related to data analysis. I was also responsible for preparing & marking two courseworks which constituted the assessment portion of the class. Additionally, I was required to be available to provide regular assistance to students.

Languages

English - Native, **Spanish** - B1, **Dutch** - A2.

Publications

Journal Papers

J7. J. M. Walker, A. Coraddu, and L. Oneto, "Decoupled Calm-water Resistance Prediction: Improving Computational Efficiency and Physical Interpretability in Hull-Form Design Optimization," *(Submitted)*, 202x

J6. A. Coraddu, J. M. Walker, and L. Oneto, "Artificial Intelligence-Based Surrogate Models for Ship Resistance and Wave Pattern Prediction during Early-Stage Ship Design," *(Submitted)*, 202x

J5. J. M. Walker, A. Coraddu, and L. Oneto, "Computationally aware and physically plausible hull form optimization incorporating the imo intact stability code," *Ocean Engineering*, vol. 341, p. 122607, 2025

J4. J. M. Walker, A. Coraddu, and L. Oneto, "A review on shape optimization of hulls and airfoils leveraging Computational Fluid Dynamics Data-Driven Surrogate models," *Ocean Engineering*, vol. 312, p. 119263, 2024

J3. J. M. Walker, A. Coraddu, and L. Oneto, "Data-Driven Models for Yacht Hull Resistance Optimization: Exploring Geometric Parameters Beyond the Boundaries of the Delft Systematic Yacht Hull Series," *IEEE Access*, vol. 12, pp. 76102–76120, 2024

J2. A. Coraddu, L. Oneto, J. Walker, K. Patryniak, A. Prothero, and M. Collu, "Floating offshore wind turbine mooring line sections health status nowcasting: From supervised shallow to weakly supervised deep learning," *Mechanical Systems and Signal Processing*, vol. 216, p. 111446, 2024

J1. J. Walker, A. Coraddu, M. Collu, and L. Oneto, "Digital twins of the mooring line tension for floating offshore wind turbines to improve monitoring, lifespan, and safety," *Journal of Ocean Engineering and Marine Energy*, vol. 8, p. 1–16, 2022

Conference Papers 🏆 *Won a best paper award.* 💬 *Conferences where I presented.*

C7. A. Coraddu, J. M. Walker, R. Dick, and L. Oneto, "From Fouling to Fuel Savings: A Data-driven Assessment of Hull & Propeller Cleaning Effectiveness," *Proceedings of Modelling and Optimisation of Ship Energy Systems (MOSES) 2025, Genoa, Italy*, Sept. 2025

C6. J. M. Walker, A. Coraddu, and L. Oneto, "Decoupled Hull Resistance Prediction: A Computationally Aware and Physically Plausible Approach for Data-Driven Surrogates," in *Proceedings of the 12th Conference on Computational Methods in Marine Engineering, Edinburgh, Scotland*, 2025 💬

C5. J. M. Walker, A. Coraddu, and L. Oneto, "Power Demand Forecasting of a Hybrid Marine Energy System using Shallow and Deep Learning," in *IEEE OCEANS, Singapore*, 2024 💬

C4. J. M. Walker, A. Coraddu, S. Savio, and L. Oneto, "Shallow and Deep Learning Models for Vessel Motions Forecasting during Adverse Weather Conditions," *Proceedings of Modelling and Optimisation of Ship Energy Systems (MOSES) 2023, Delft, Netherlands.*, Dec. 2023 💬 🏆

C3. J. M. Walker, A. Coraddu, and L. Oneto, "A Decoupled Approach to AI-based Design and Optimization of the Delft Systematic Yacht Hull Series," in *Conference on Computer and IT Applications in the Maritime Industries*, pp. 209–222, 2023 💬

C2. J. M. Walker, A. Coraddu, V. Garofano, and L. Oneto, "Artificial Intelligence Based Short-Term Motions Forecasting for Autonomous Marine Vehicles Control," in *Proceedings of the International Ship Control Systems Symposium*, vol. 16, p. 35, 2022

C1. J. Walker, A. Coraddu, L. Oneto, and S. Kilbourn, "Digital Twin of the Mooring Line Tension for Floating Offshore Wind Turbines," in *IEEE OCEANS, San Diego–Porto*, 2021

Book Chapters

BC1. A. Coraddu, M. Kalikatzarakis, J. Walker, D. Ilardi, and L. Oneto, "Data science and advanced analytics for shipping energy systems," in *Sustainable Energy Systems on Ships*, pp. 303–349, Elsevier, 2022